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CLASS: BSAI **SECTION: 3A**

LAB TASK NO: 3

TASK :1

Tracing BFS & DFS

Graph:

A -> B, D, C

B -> E

D -> F

C -> F

E -> C

F -> G

BFS (FIFO):

Steps Visited Queue

1 A B,D,C

2 B D,C,E

3 D C,E,F

4 C E,F

5 E F

6 F G

7 G []

Order: A -> B -> D -> C -> E -> F -> G

Path: A -> D -> F -> G

DFS (LIFO):

Order: A -> B -> E -> C -> F -> G

Path: A -> B -> E -> C -> F -> G

Task 2:

DLS & IDDFS

Graph:

A -> B, C

B -> D, E

C -> F

E -> G

Goal: G

```
def DLS(graph, start, goal, limit, path):
if start == goal:
        return path
if limit <= 0:
        return 'CUTOFF'
cutoff = False
for child in graph[start]:
        r = DLS(graph, child, goal, limit - 1, path + [child])
 if r == 'CUTOFF':
            cutoff = True
return r if not cutoff else 'CUTOFF'
def IDDFS(graph, start, goal, max_depth):
    for d in range(max_depth + 1):
r = DLS(graph, start, goal, d, [start])
if r != 'CUTOFF':
            return r
```

DLS Output:

Depth Result

Task 3:

Uniform Cost Search

Uniform cost search will always expand the path with the least total cost.

Edges:

A-B = 11

A-D = 7

A-C = 14

B-E = 15

E-C = 8

C-F = 10

D-F = 25

F-G = 20

E-H = 9

H-G = 10

```
import heapq


graph = {
    'A': [('B', 11), ('D', 7), ('C', 14)],
    'B': [('E', 15)],
    'D': [('F', 25)],
    'C': [('F', 10)],
    'E': [('C', 8), ('H', 9)],
    'F': [('G', 20)],
    'H': [('G', 10)],
    'G': []
}
```

```
def ucs(start, goal):

    queue = [(0, start, [start])]
    visited = set()

    while queue:

        cost, node, path = heapq.heappop(queue)

        if node in visited:
            continue

        visited.add(node)

        print('Visited Node:', node, 'Cost:', cost)

        if node == goal:
            return path, cost

        for child, w in graph[node]:

            if child not in visited:
                heapq.heappush(
                    queue,
                    (cost + w, child, path + [child]))
```

return [], 0

path, cost = ucs('A', 'G')

print('Final Path:', path)

print('Total Cost:', cost)

Visited Nodes:

Step Node Cost

1 A 0

2 D 7

3 B 11

4 C 14

5 F 24

6 E 26

7 H 35

8 G 44

UCS Path: A -> C -> F -> G (Total Cost: A-D = 70__)

Compare BFS vs UCS:

Algorithm Path Cost

BFS A->D->F->G 52

UCS A->C->F->G 44

GITHUB LINK:

https://github.com/umair-900/Ai-lab-Task